Objective:

The objective of this lab is to design a Differential to a single ended amplifier and measure the Rejection ratios (CMRR and PSRR).

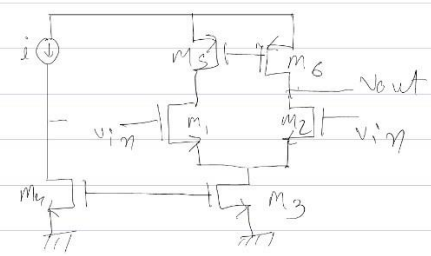
Process Parameters:

- NMOS: tsmc25N: Kn' = 2.708×10-4 A/V2
- PMOS: tsmc25P: Kp' = 0.944×10-4 A/V2
- Vdd = 1.6V, total current from supply: IVDD= 33 dB.
- Output swing >= 1.2 Vpp. BW>2.5 MHz CL = 1pF

Hand Calculations:

NMOS = $K_n' = 2.708 \times 10^{-4}\ A/V^2$

PMOS = $K_p' = 0.944 \times 10^{-4}\ A/V^2$



Calculating width of transistors M_1 and M_3

$$I = \frac{K_n' W_6}{2 L}(V_{ov})^2 \quad , \quad L = 600nm$$

$$7.5 \times 10^{-6} = \frac{2.708 \times 10^{-4}}{2} \times \frac{W}{600 \times 10^{-9}} \times (10^{-2})$$

$$W = 33.22\ \mu m$$

Now, Calculating width of transistor M_3

$$I = \frac{K_n'}{2} \frac{W}{L} (V_{ov})^2$$

$$150 \times 10^{-6} = \frac{2.708 \times 10^{-4}}{2} \times \frac{W}{600 \times 10^{-9}} \times 10^{-2}$$

$$W = 65.4 \, \mu m$$

Calculating width of transistor m_4

$$I = \frac{K_n'}{2} \frac{W}{L} (V_{ov})^2$$

$$10 \times 10^{-6} = \frac{2.708 \times 10^{-4}}{2} \times \frac{W}{600 \times 10^{-9}} \times 10^{-2}$$

$$W = 4.431 \, \mu m$$

Calculating width of transistor m_5 and m_6

$$I = \frac{K_p'}{2} \frac{W}{L} (V_{ov})^2$$

$$7.5\times10^{-6} = \frac{0.9444\times10^{-9}}{2} \times \frac{W}{600\times10^{-9}} \times 10^{-2}$$

$$W = 95.32\ \mu m$$

$$\text{Diffrential gain} = \frac{g_m}{1+g_m}\left[(r_o + g_m r_o)\right]$$

$$g_m = \frac{2\times I_D}{V_{OV}}$$

$$= \frac{2\times10\times10^{-6}}{0.01} = 2\times10^{-3}$$

$$g_m = 2\,mS$$

$$r_o = \frac{1}{\lambda I_D}$$

$$= \frac{1}{0.127\times10\times10^{-6}} = 787\ K\Omega$$

$$\text{gain}(A_V) = \frac{2\times10^{-3}}{1+2\times10^{-3}}\left(787\times10^{3} + 2\times10^{-3}(787\times10^{3})\right)$$

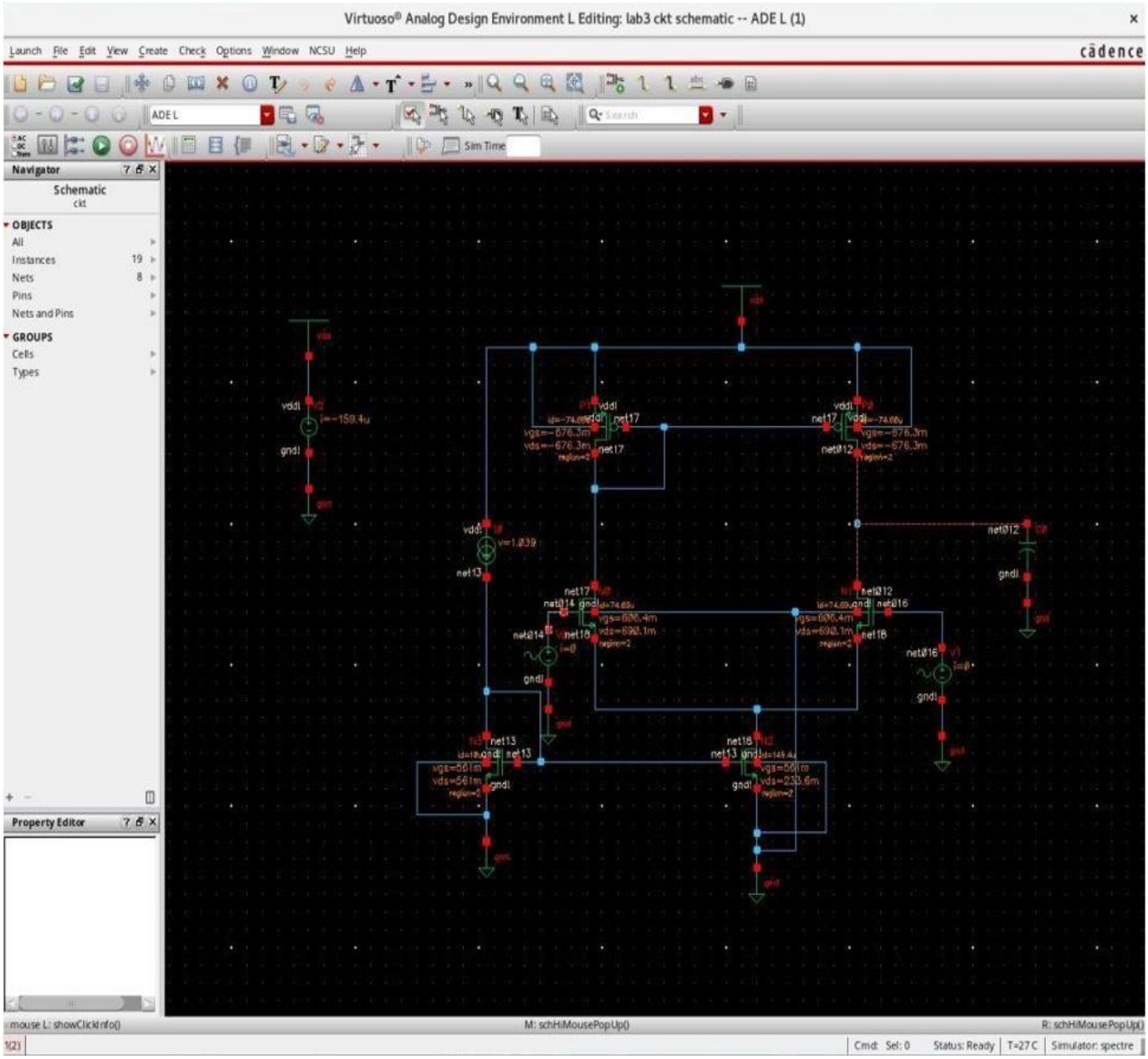
$$A_V = 62.4\ V/V$$

$$A_V(db) = 20\log_{10}(62.4)$$

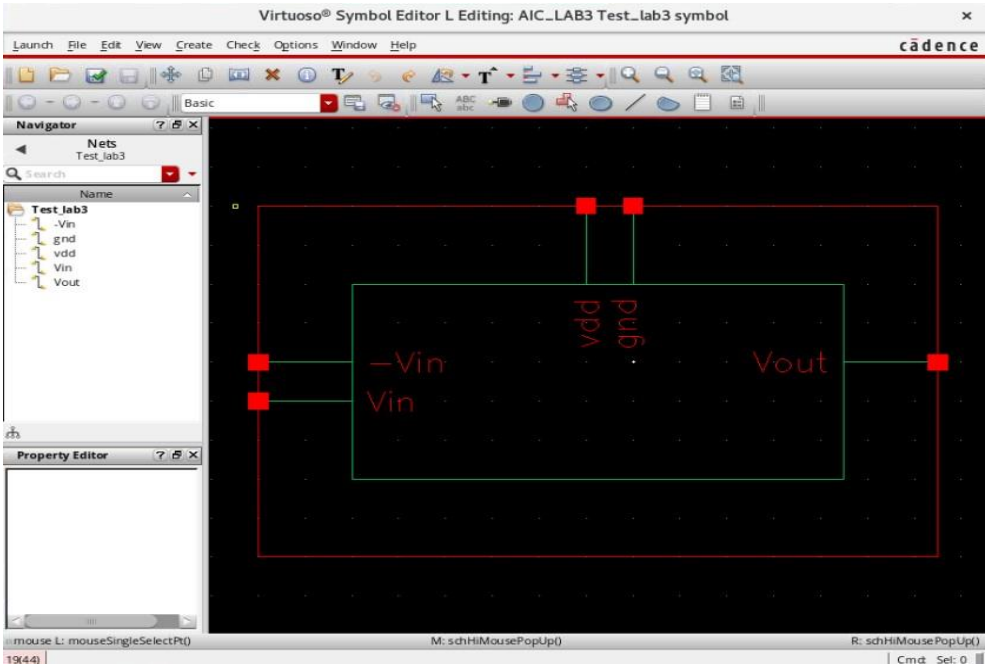
$$A_V(db) = 35.903$$

Design Analysis:

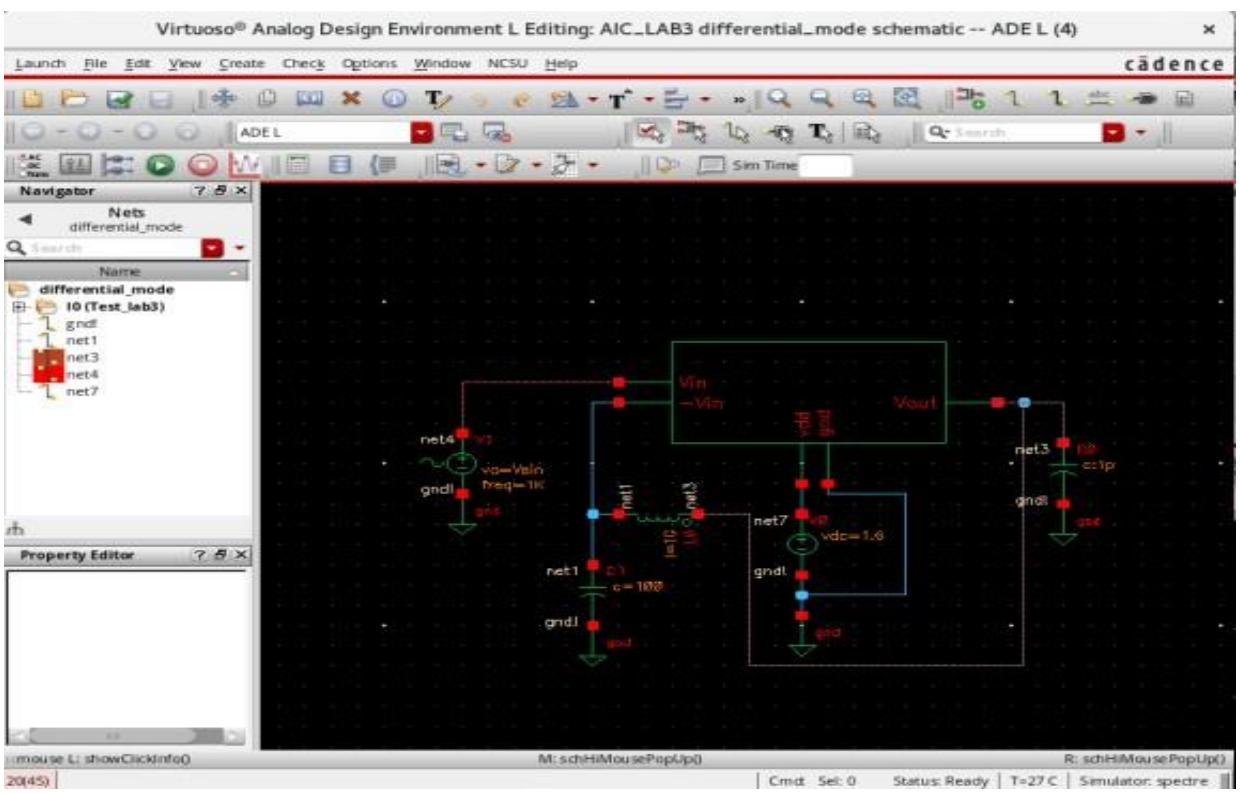
DC parameters:



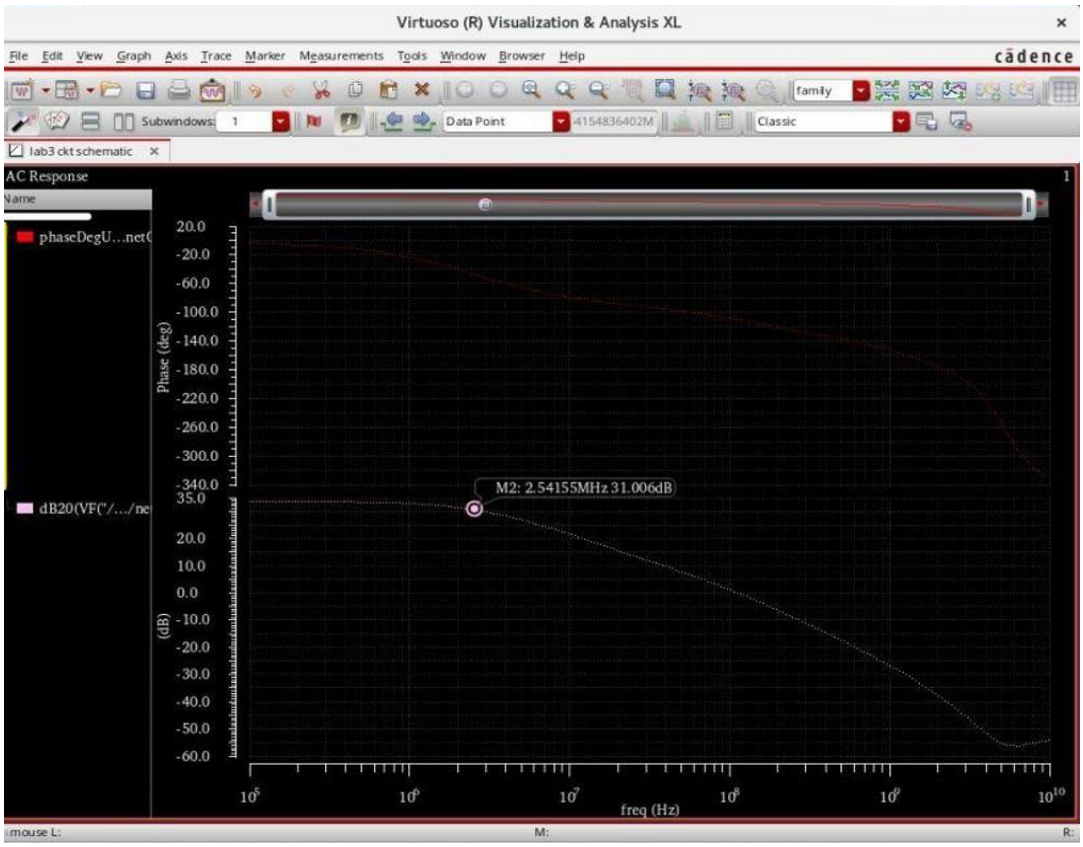
Symbol



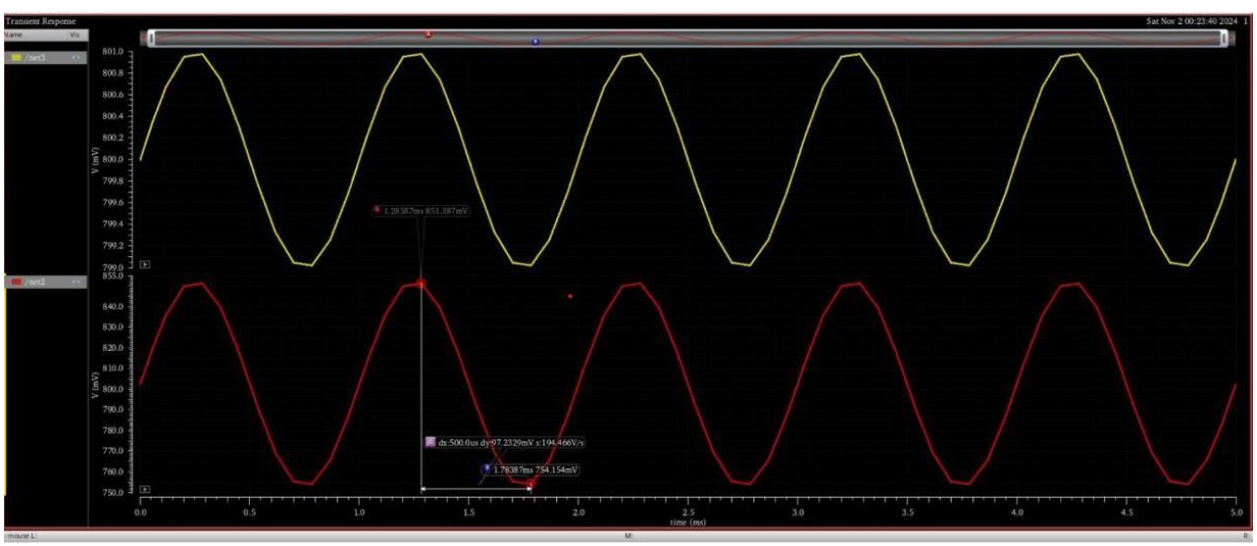
Differential mode circuit diagram:

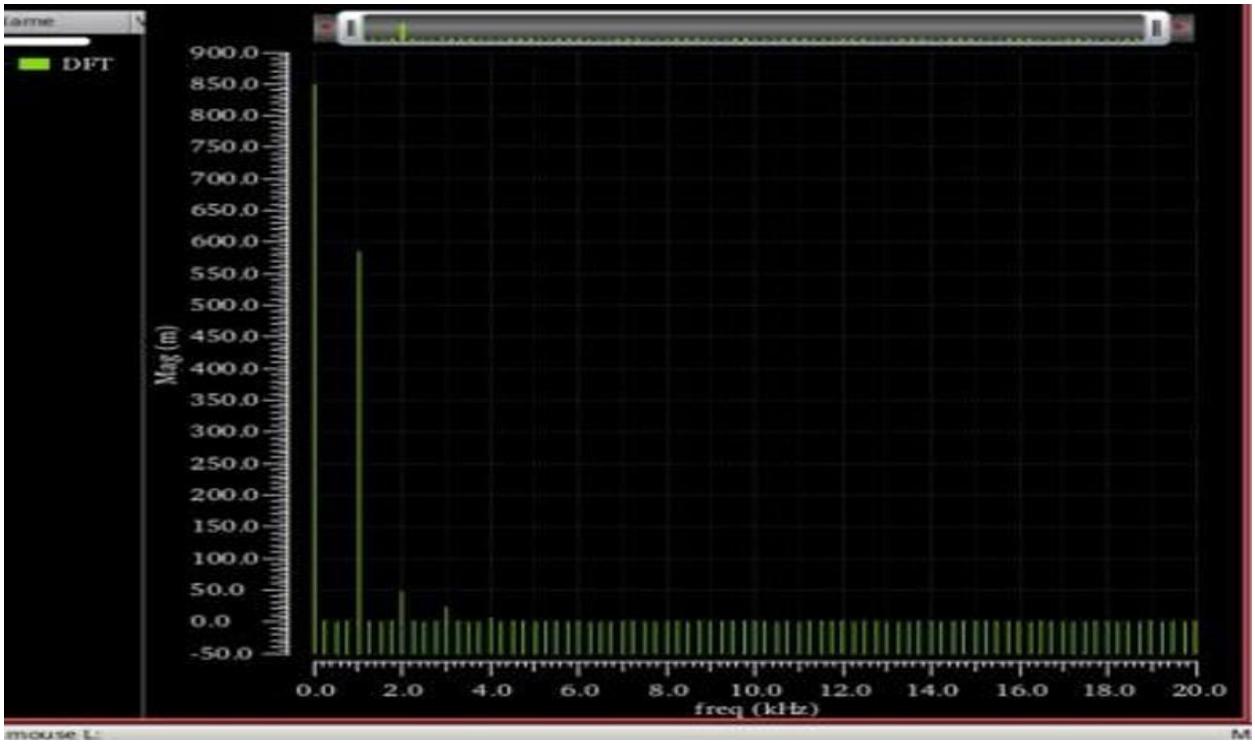


Gain and Phase:

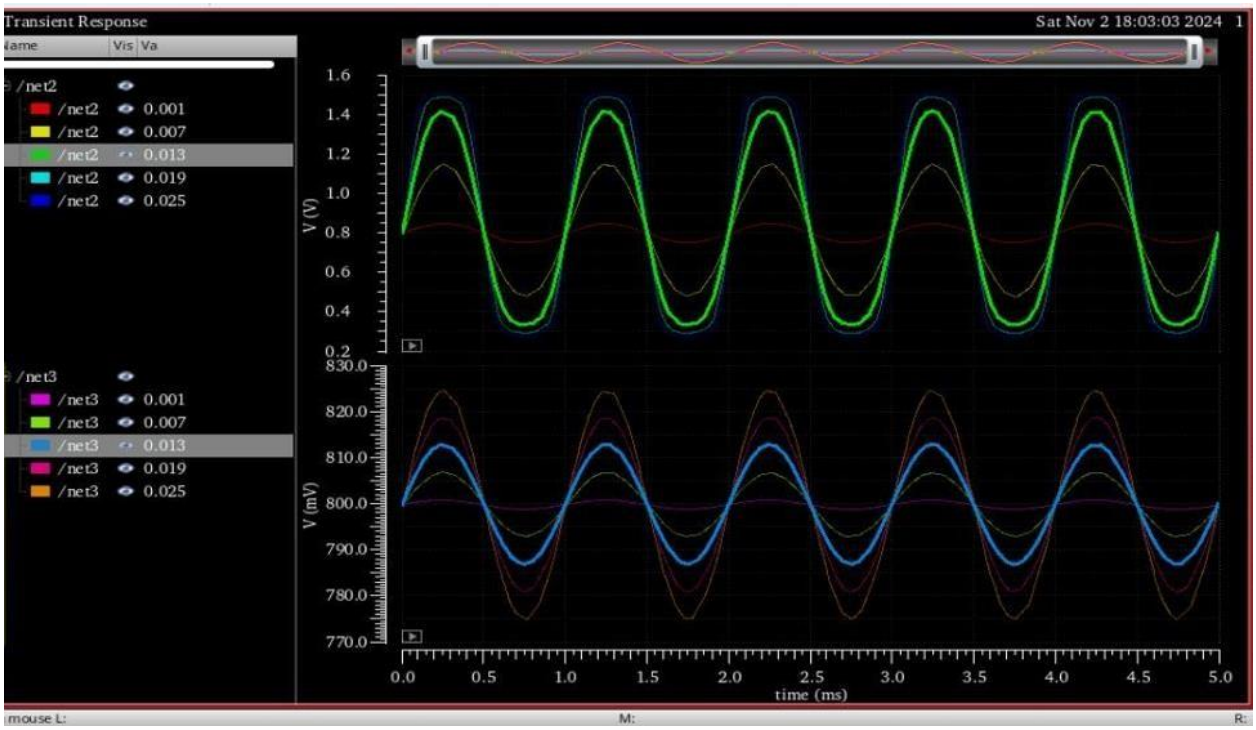


Gain and Phase VS Frequency**: AC Gain = 31.006dB, BW= 2.54155 MHz**





Spectrum Plot

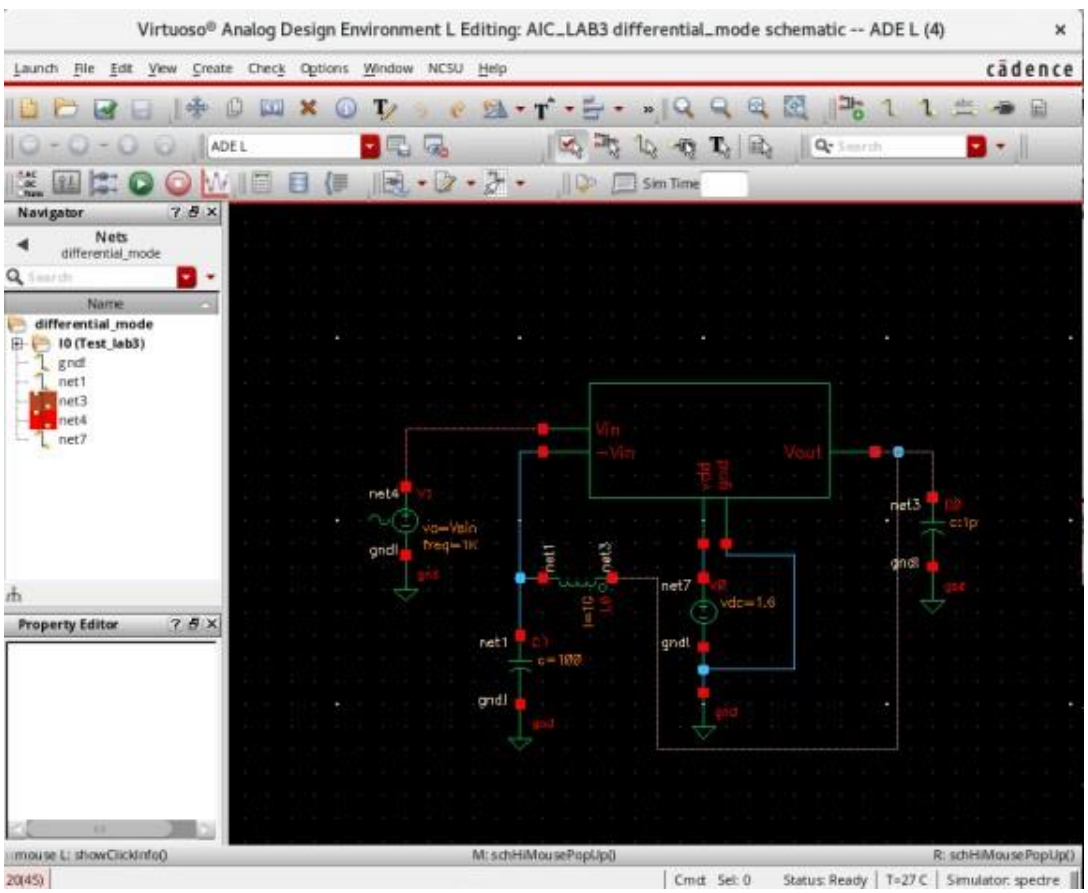


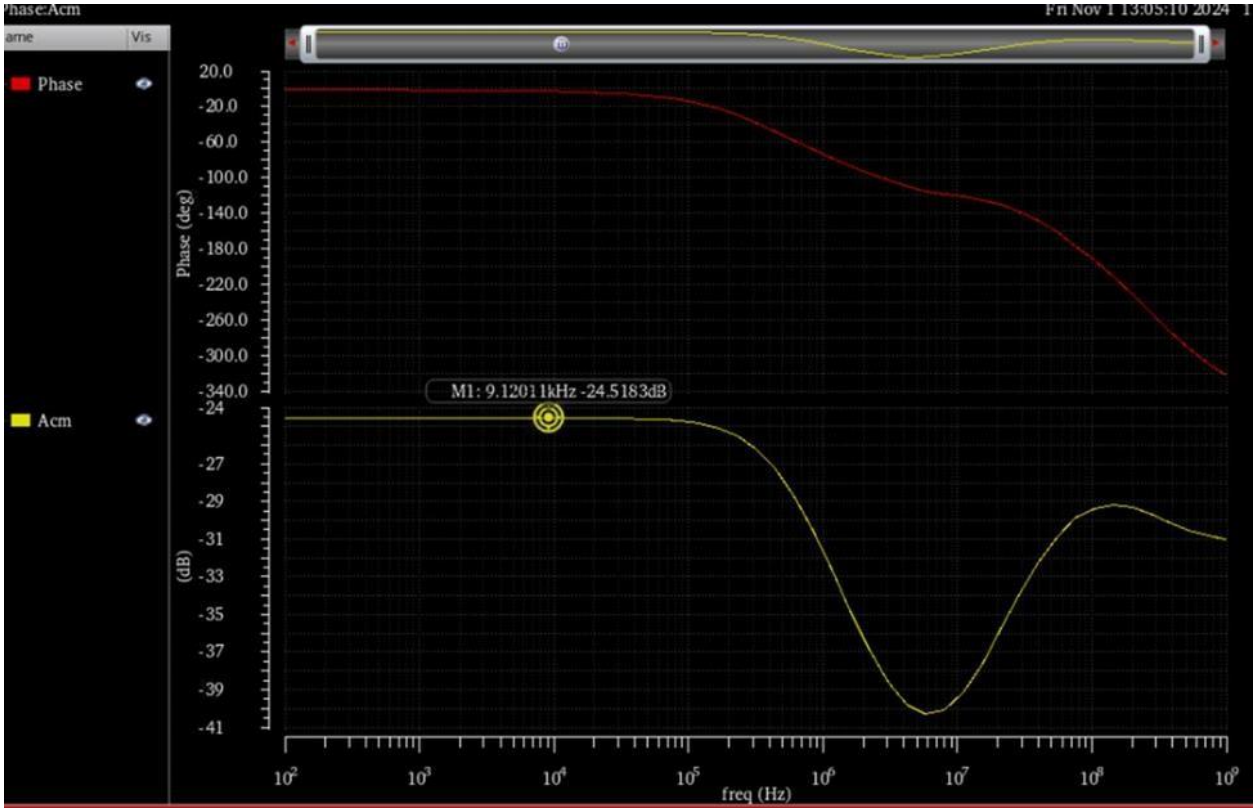
Transient Response

Parameter	Simulated value	Hand Value
Width	NMOS (m1,m2)- 33.23um NMOS m3- 66.525um, m4-4.45um. PMOS (m5,m6) – 95.34um	NMOS (m1,m2)- 33.221um NMOS m3- 65.452um, m4-4.425um. PMOS (m5,m6) – 95.3 um
Vin	805mV	0.8V
Gain	34.2dB	-39.9dB
Length	600nM	600nM

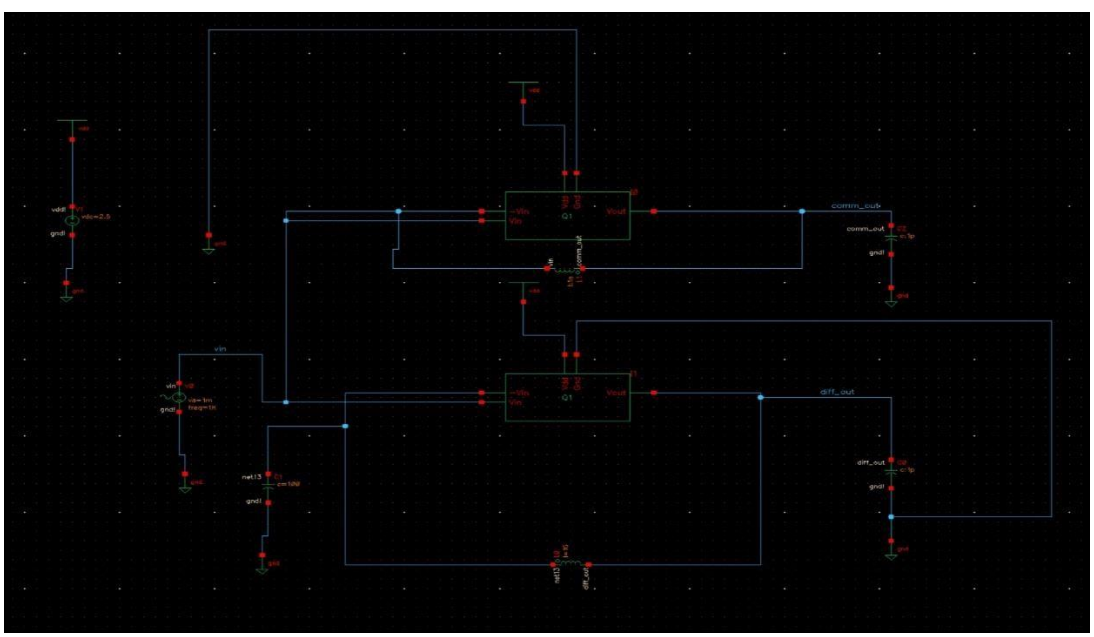
Output Swing with Differential Transient Input: This plot illustrates the response of the amplifier's output to a quickly changing 30 mV differential input signal.

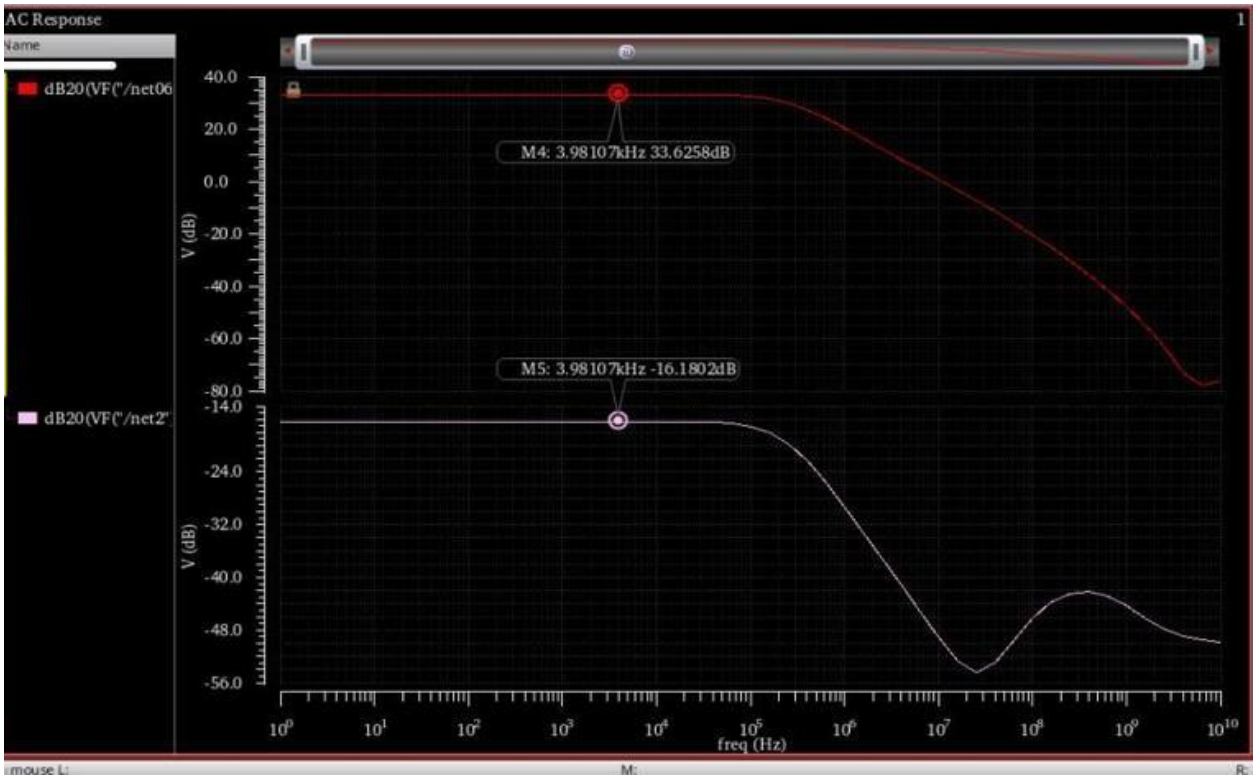
Common Mode:

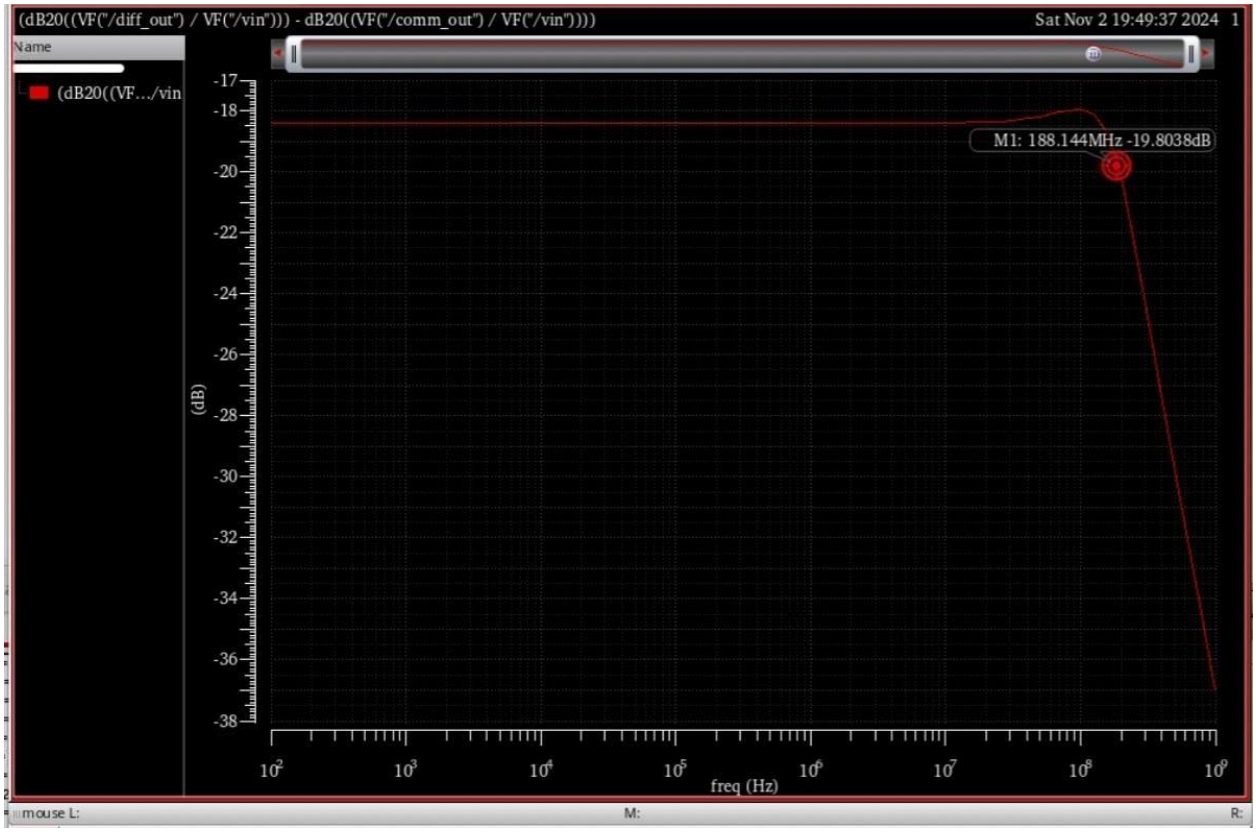




CMRR Schematic




AC Response
Name
dB20(VF("/net06
dB20(VF("/net2"
M4: 3.98107kHz 33.6258dB
M5: 3.98107kHz -16.1802dB
V (dB)
freq (Hz)


(dB20((VF("/diff_out") / VF("/vin"))) - dB20((VF("/comm_out") / VF("/vin"))))
Sat Nov 2 19:49:37 2024
Name
(dB20((VF.../vin
M1: 188.144MHz -19.8038dB
(dB)
freq (Hz)

= -19.8038 dB which is defined as Ad/Acm.

CMRR = (dB20((VF("/Differential_mode_Output") / VF("/Input"))) - dB20((VF("Common_mode_Output") / VF("/Input"))))

When there is no offset at input Output DC Voltage = $\frac{V_{DD}}{2} = 0.8V$

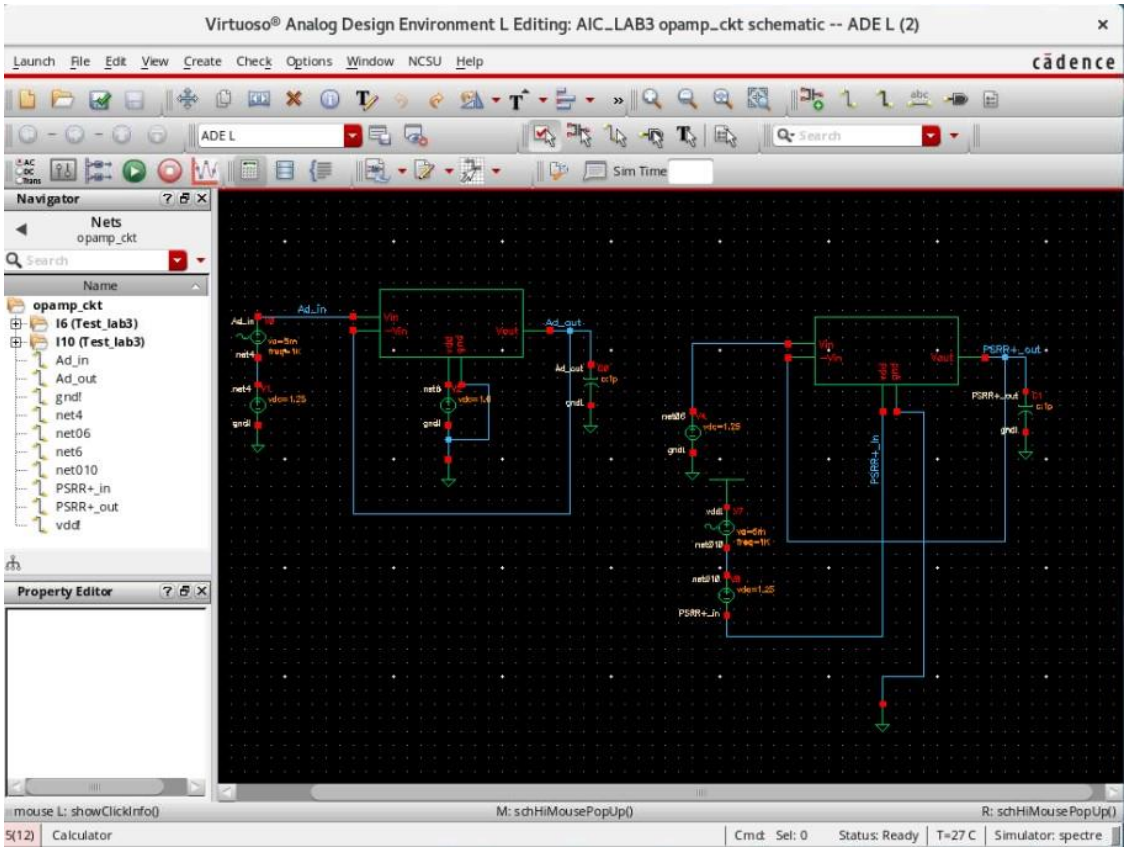
When there is a 5 mV offset
New DC Output Voltage
$= (V_{DD}/2) + (A_d)(5mV)$

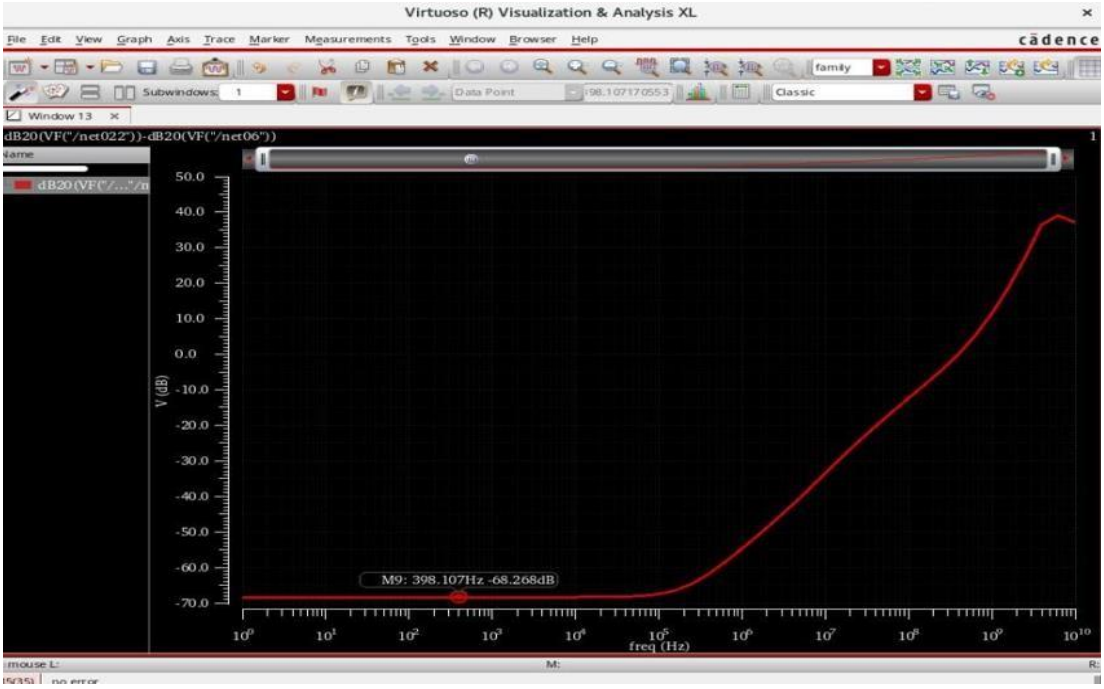
$A_d = 34.0dB$

$\therefore V_{out\,DC} = 0.8 + (34dB) \times 5 \times 10^{-3}$
$= 0.8 + (250.59 \times 10^{-3})$
$\Rightarrow 1.05V$

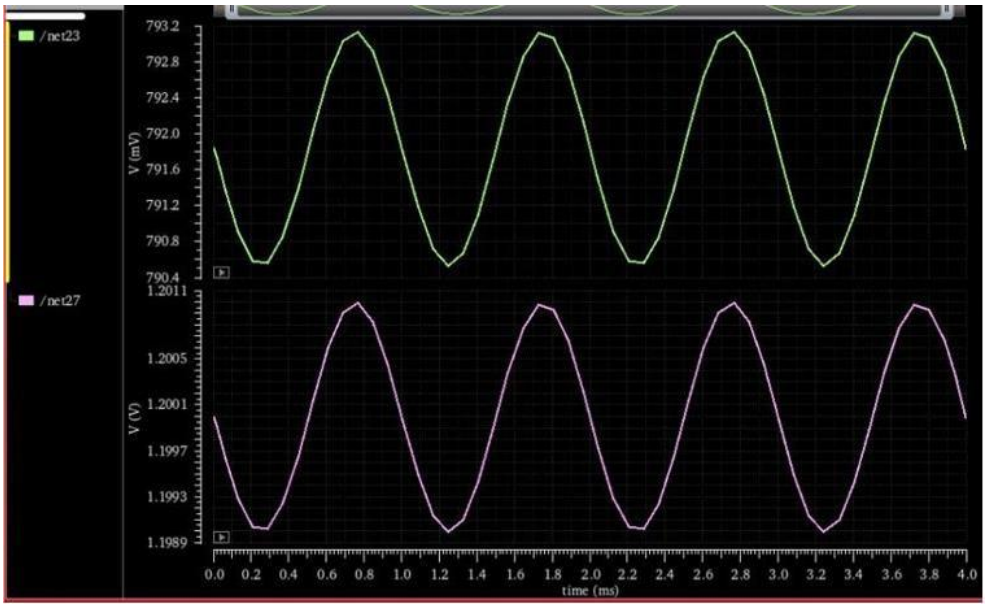
Assessing how an amplifier responds to common-mode signals is crucial for determining how effective it is in real-world situations. We can evaluate the amplifier's ability to reduce undesired noise by looking at metrics such output common-mode voltage (VOCM), common-mode gain, and Common Mode Rejection Ratio (CMRR). With this knowledge, we can put policies in place that reduce common-mode interference and encourage consistent performance in a variety of applications.

<u>Supply Rejection :</u>

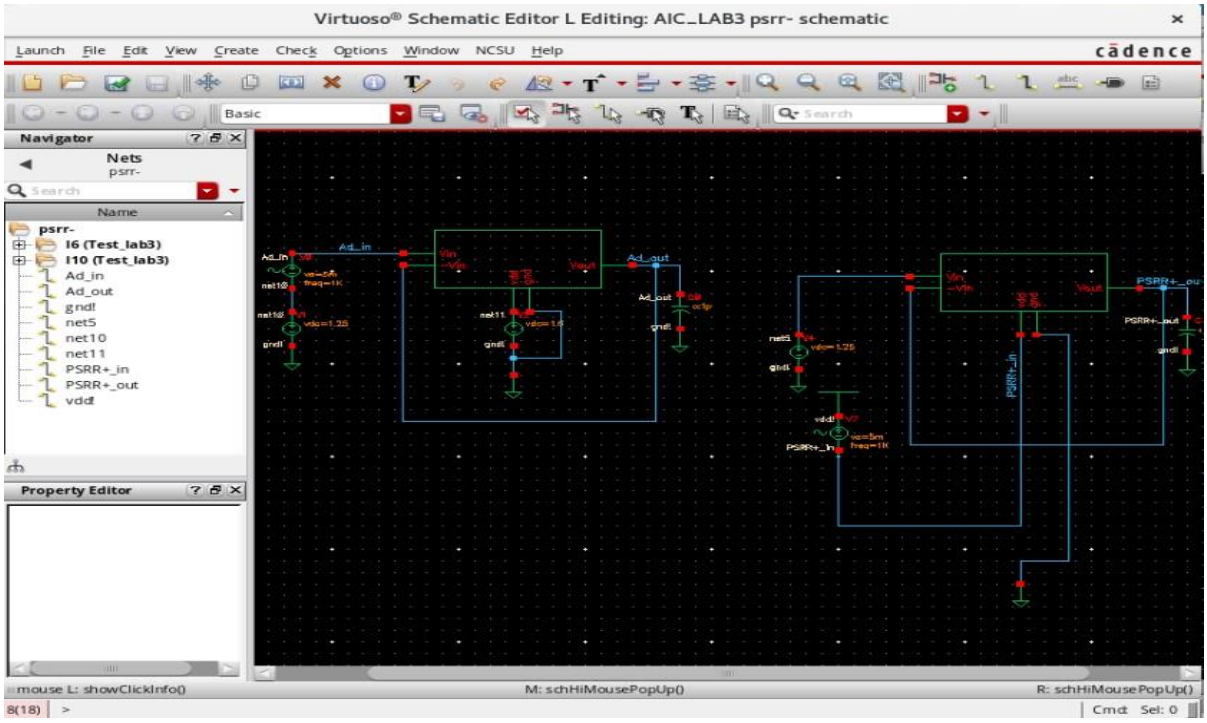

Virtuoso® Analog Design Environment L Editing: AIC_LAB3 opamp_ckt schematic -- ADE L (2)
Launch File Edit View Create Check Options Window NCSU Help
cādence
ADE L
Sim Time
Navigator
Nets
opamp_ckt
Name
opamp_ckt
I6 (Test_lab3)
I10 (Test_lab3)
Ad_in
Ad_out
gnd!
net4
net06
net6
net010
PSRR+_in
PSRR+_out
vdd!
Property Editor
Ad_in
Ad_out
PSRR+_out
PSRR+_in
mouse L: showClickInfo()
M: schHiMousePopUp()
R: schHiMousePopUp()
Calculator
Cmd: Sel: 0
Status: Ready
T=27 C
Simulator: spectre


Virtuoso (R) Visualization & Analysis XL
File Edit View Graph Axis Trace Marker Measurements Tools Window Browser Help
cādence
Subwindows: 1
Data Point
Classic
Window 13
dB20(VF("/net022"))-dB20(VF("/net06"))
V (dB)
50.0
40.0
30.0
20.0
10.0
0.0
-10.0
-20.0
-30.0
-40.0
-50.0
-60.0
-70.0
M9: 398.107Hz -68.268dB
freq (Hz)
mouse L:
M:
R:

PSRR Frequency

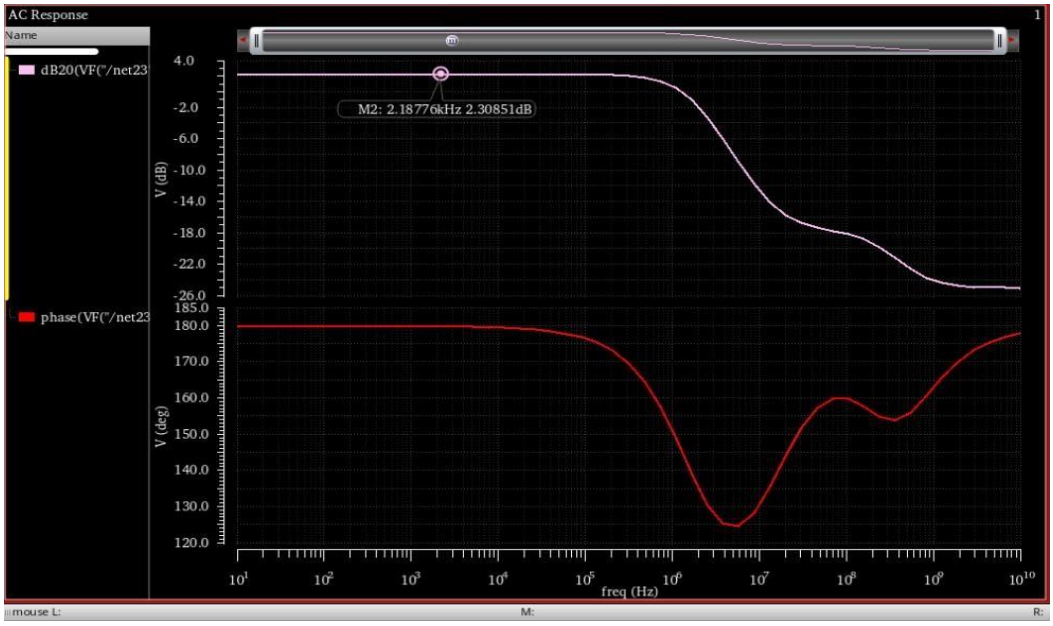


Transient Analysis for PSRR+ and PSRR_Out

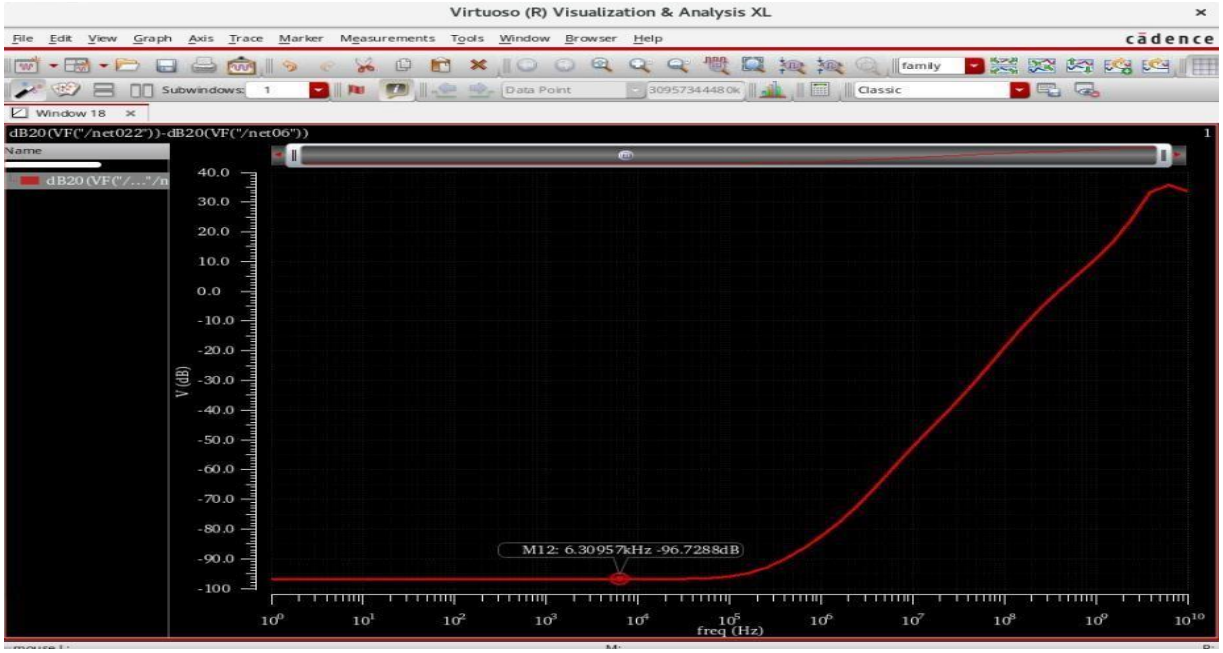


PSRR- circuit design

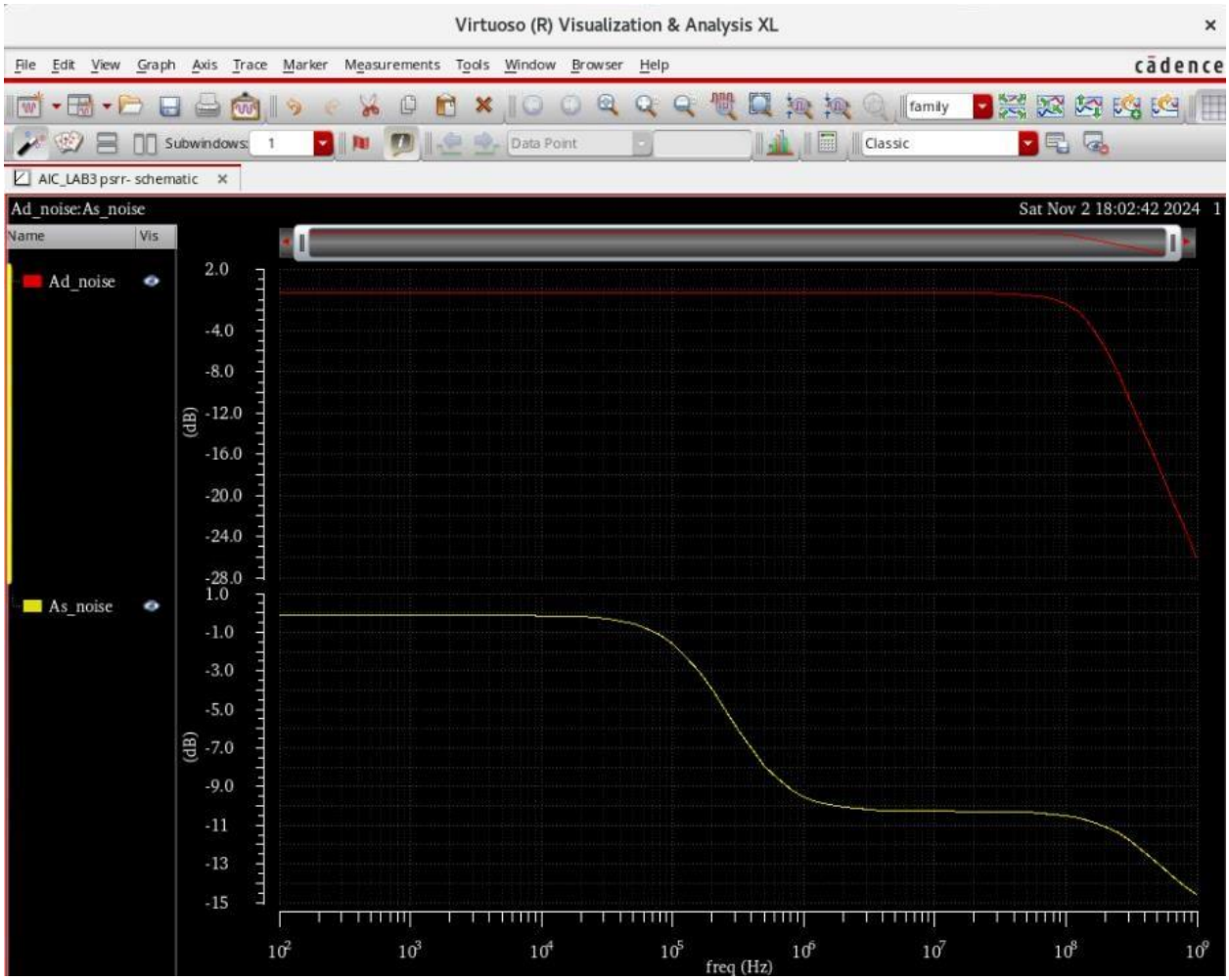
Noise Gain (As_noise) V/S Frequency



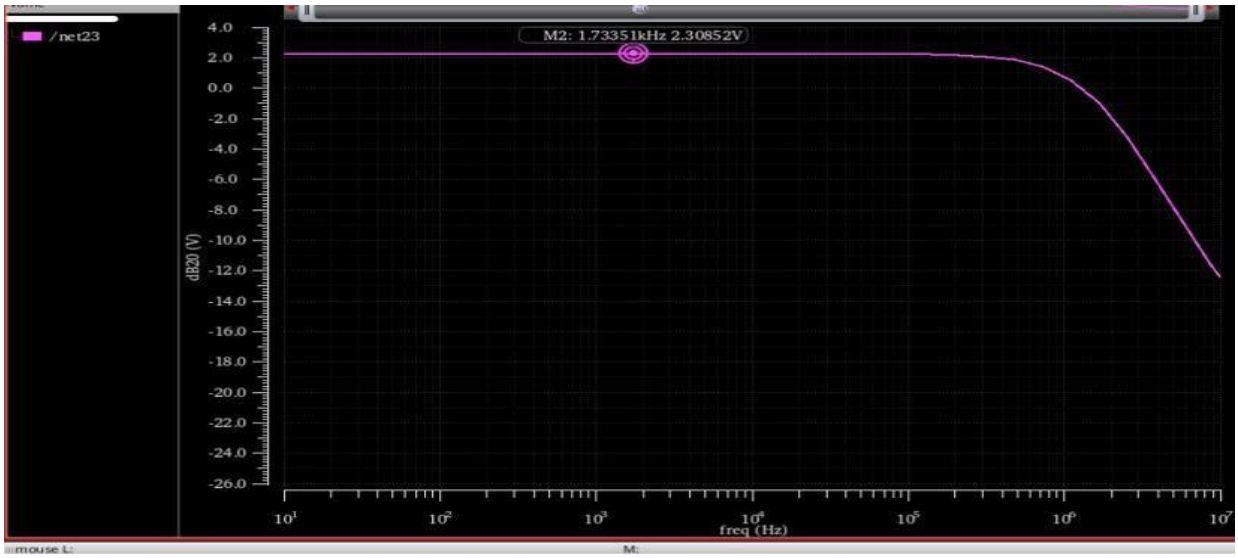
Noise Gain (As Noise) VS Frequency



PSRR frequency



AD gain



PSRR = As_noise – Ad (in dB)

PSRR = (dB20((VF("/PSRR+_Out") / VF("/PSRR+_In"))) - dB20((VF("Diff_Out") / VF("/Diff_In"))))

• The hand-calculated values and the simulations displayed are almost equal; the differential amplifier is developed and simulated for the specified parameters.

•When the simulation and manually calculated values of several parameters were compared to the observed values, they appeared to be almost identical.